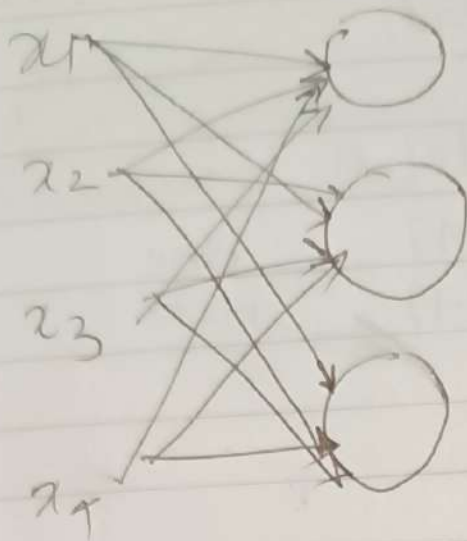
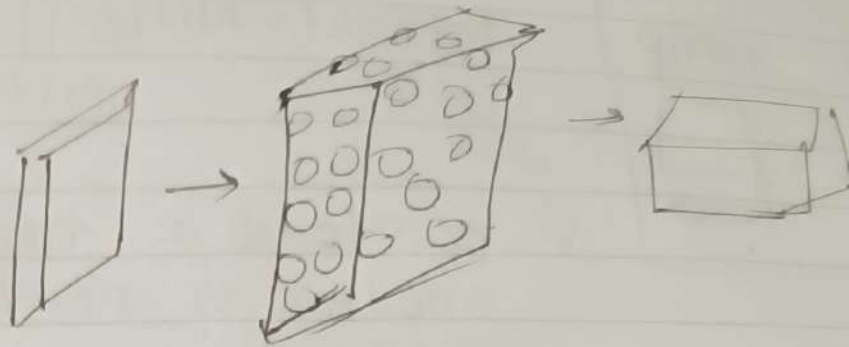


Input Output

Neural Network example



Standard NN



CNN

Convolutional

Application ML

Unstructured data

Structured data

Size	bedroom	Price
2104	3	400
1600	3	300
.	.	.

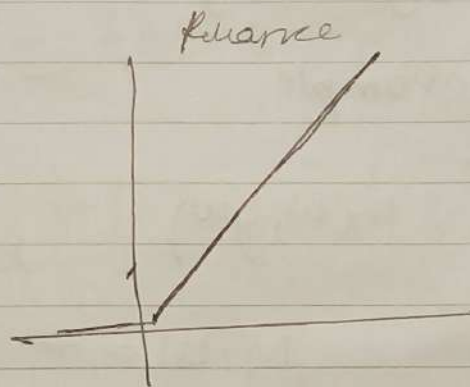
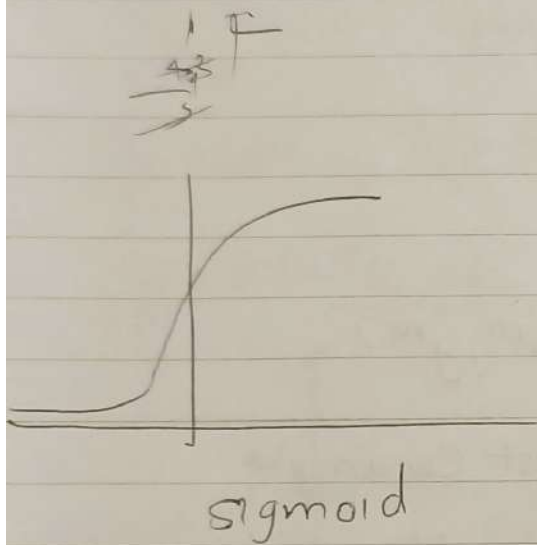
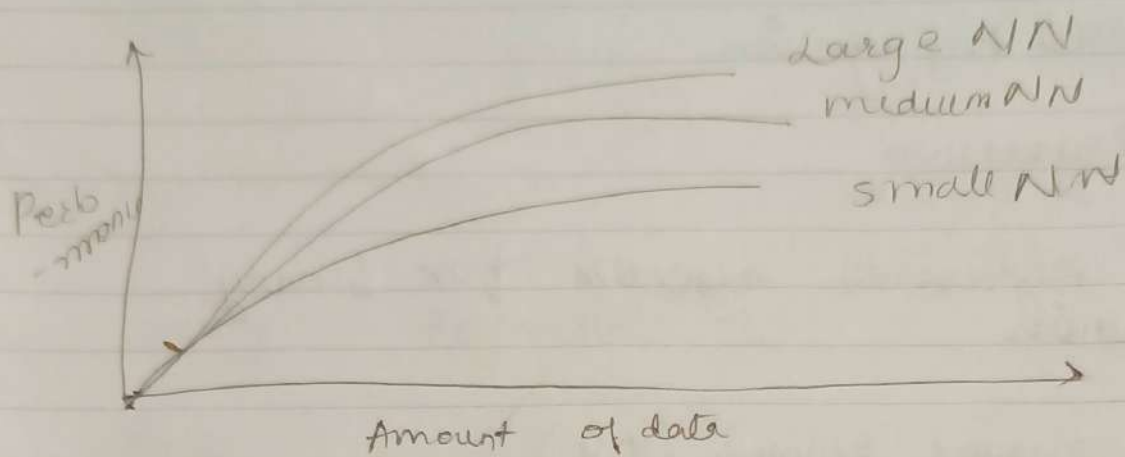
Audio

Image → Pixel

Text

Structured format

scale drives deep learning power



Neural Networks and deep learning

Dull

Introduction

Basis of Neural Network Programming

~~Week 3~~ One hidden layer

deep Neural Network

forward propagation

Backward propagation

Logistic Regression

→ Logistic Regression algorithm for binary classification

single training example (x, y)

$$x \in \mathbb{R}^n, y \in \{0, 1\}$$

m training example

$$(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})$$

$M = M_{\text{training}}$

$M_{\text{test}} = \# \text{ test example}$

$$X = \begin{bmatrix} | & | & | \\ x^{(1)} & x^{(2)} & \dots & x^{(m)} \\ | & | & | \end{bmatrix} \quad \begin{matrix} \uparrow \\ n \times \\ \downarrow \end{matrix}$$

$$X \in \mathbb{R}^{n \times m}$$

$$X.\text{shape} = (n, m)$$

$$y = [y^{(1)}, y^{(2)}, \dots, y^{(m)}]$$
$$y \in \mathbb{R}^{1 \times m}$$

$$y.\text{shape} = (1, m)$$

$$X = \begin{bmatrix} | & | & | \\ x^{(1)} & x^{(2)} & \dots & x^{(m)} \\ | & | & | \end{bmatrix}$$

$$Y = [y^{(1)}, y^{(2)}, \dots, y^{(m)}]$$

$$y \in \mathbb{R}^{(1 \times m)}$$

$$X \text{ shape} = (n \times m)$$

$$Y \text{ shape} = (1, m)$$

Logistic Regression

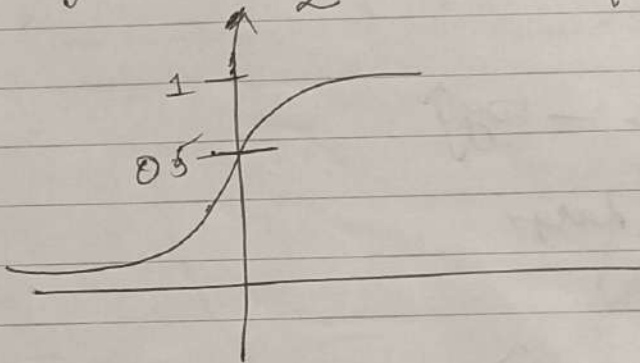
Given x want $\hat{y} = P\left(\frac{y=1}{x}\right)$

$$x \in \mathbb{R}^n$$

Parameters $w \in \mathbb{R}^n$ $b \in \mathbb{R}$

$$y = \sigma\left(\frac{w^T x + b}{z}\right)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$



If z very large,
 $-z = 0$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\frac{e^z = 0}{1 + 0} = 1$$

$$= \frac{1}{1 + \text{very very big}}$$

$$\hat{y} = \sigma(\theta^T x)$$

$$\theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_{n-1} \end{bmatrix} \quad \left. \begin{matrix} \theta_0 \\ \theta_1 \\ \theta_2 \\ \vdots \\ \theta_{n-1} \end{matrix} \right\} w$$

Logistic Regression model

$$\hat{y} = \sigma(w^T x + b) \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Given $\{x^{(1)}, y^{(1)}\} \dots \{x^{(m)}, y^{(m)}\}$

$$Z(i) = w^T x(i) + b$$

Loss function error

$$\mathcal{L}(\hat{y}, y) = \frac{1}{2} (y - \hat{y})^2$$

$$\mathcal{L}(\hat{y}, y) = -(y \log \hat{y} + (1-y) \log (1-\hat{y}))$$

If $y = 1$ $\mathcal{L}(\hat{y}, y) = -\log \hat{y}$... want $\log \hat{y}$ Large
 $\rightarrow \hat{y}$ Large

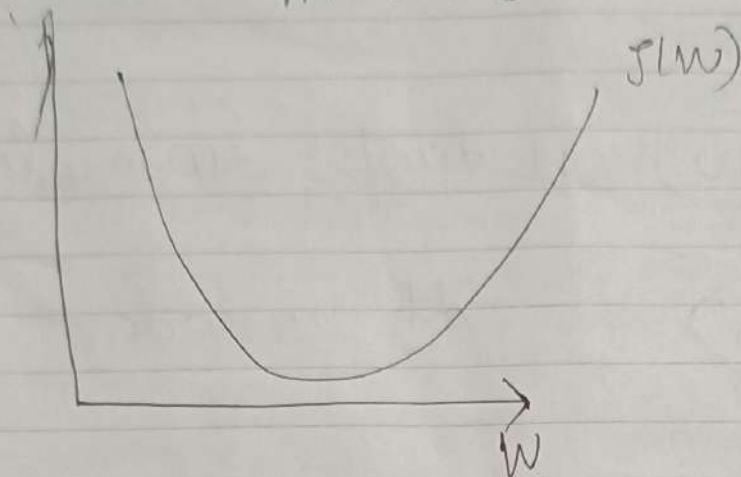
~~$\log \hat{y}$~~ If $y = 0$ $\mathcal{L}(\hat{y}, y) =$

$\mathcal{L}(\hat{y}, y) = -\log (1-\hat{y}) \leftarrow$ want $\log (1-\hat{y})$ Large
 $\rightarrow$ want $\hat{y}$ small

Cost function major cause

$$\text{cost function } J(W, b) = \frac{1}{m} \sum_{i=1}^m L(y^{(i)}, \hat{y}^{(i)})$$

$$= \frac{1}{m} \sum_{i=1}^m [y^{(i)} \log \hat{y}^{(i)} + (1 - y^{(i)}) \log (1 - \hat{y}^{(i)})]$$



Repeat {

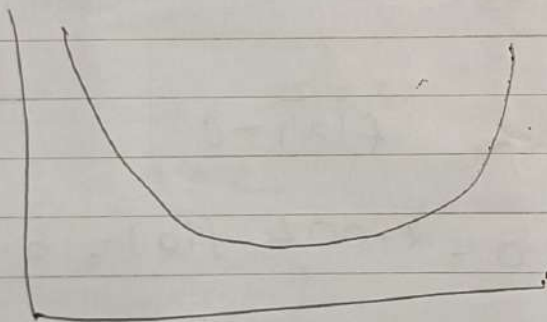
$$W_0 = W - \alpha \frac{dJ(W)}{dW}$$

learning rate

gradient descent

$$W := W - \alpha \frac{dJ(W)}{dW}$$

$$W_0 = W - \alpha dW$$



to $J(W, b)$

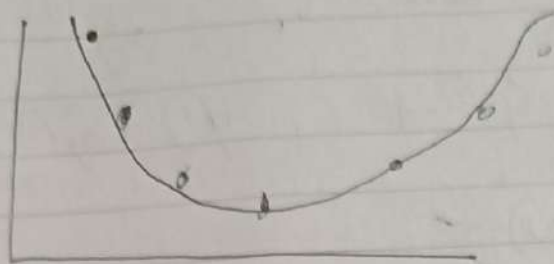
$$W_0 = W - \alpha \frac{dJ(W, b)}{dW}$$

$$b = b - \alpha \frac{dJ(W, b)}{db}$$

$$\frac{\partial J(W, b)}{\partial W}$$

partial derivative

Gradient Descent

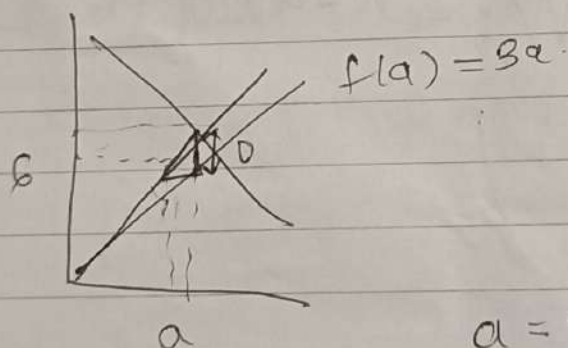


$\frac{\partial J(w, b)}{\partial w} \rightarrow dw$ it denoted dw in code

$\frac{\partial J(w, b)}{\partial b} \rightarrow db$ db in code

* Calculus Derivatives

Test or function couples function that will unable to encapsulate



$$a = 2$$

$$f(a) = 6$$

$$a = 2.001 \quad f(a) = 6.003$$

Slope of derivative of $f(a)$ at 2 = 3

$$\text{slope} = \frac{\text{height}}{\text{width}} = \frac{0.003}{0.001} = 3$$

$$a = 5$$

$$f(a) = 3a$$

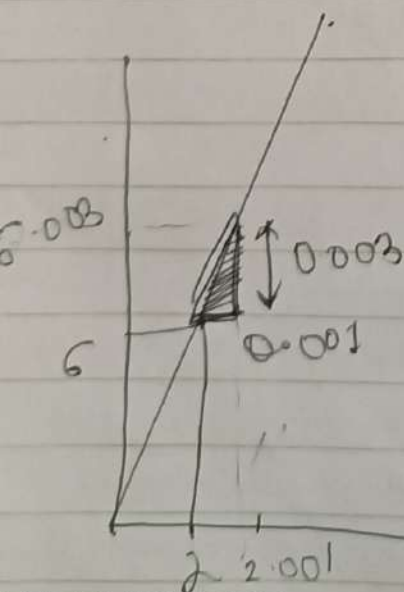
$$= 15$$

$$a = 5.001$$

$$f(a) = 15.003$$

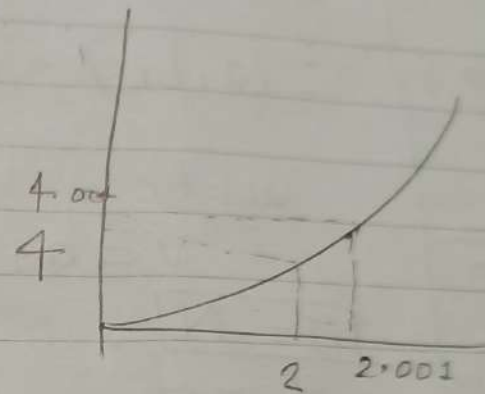
slope at $a=5$

is also 3



$$\frac{d f(a)}{da} = 3$$

$$\frac{d(f(a))}{da}$$



$$f(a) = a^2$$

$$a = 2$$

$$f(a) = 4$$

$$f(a) = 2.001 \quad f(a) = 4.004$$

Slope derivative of $f(a)$ at $a=2$

$$\frac{d f(a)}{da} = 4$$

$$f(a) = a^2$$

$$f'(a) = 2a$$

when $a=2$ $f'(a) = 4$

$$a=5 \quad f(a) = 25$$

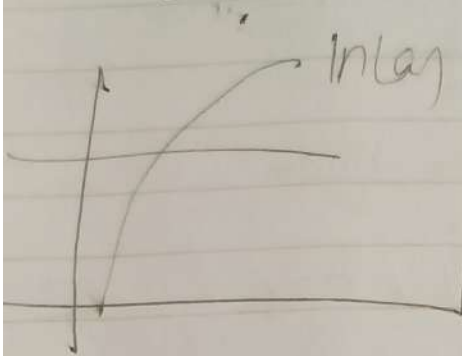
$$a=5 \quad f(a) = 25$$

$$a=5.001 \quad f(a) = 25.0010$$

computation graph

$$f(a) = \log_e(a)$$

$$\frac{df(a)}{da} = \frac{1}{a}$$



$$J(a, b, c) = 3(a + bc)$$

$$u = bc$$

$$v = a + u$$

$$J = 3v$$

$$a = 5$$

$$b = 3$$

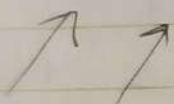
$$c = 2$$

$$u = bc$$

$$v = a + u$$

$$\begin{aligned} &= 3(5 + 3 \times 2) \\ &= 3(5 + 6) \\ &= 3 \times 11 \\ &= 33 \end{aligned}$$

$$J = 33$$



Derivative with computation

$$J = 3v$$

$$\frac{dJ}{dv} = 3$$

$$f(a) = 3a$$

$$\frac{df(a)}{da} = \frac{\partial f}{\partial a} = 3$$

$$\begin{aligned} J &= 3v \\ \frac{dJ}{dv} &= 3 \end{aligned}$$

$$a = 5 \quad 5.001$$

$$v = 11 \quad 11.001$$

$$J = 33 \quad 33.003$$

$$\frac{dJ}{dv} = 3$$

$$\frac{dJ}{dv}$$

$$\frac{dJ}{da} = \frac{dJ}{dv} \frac{dv}{da}$$

$$\frac{dJ}{da} =$$

$$\frac{dv}{da} = 1$$

$$\frac{d \text{ Final Output}}{d \text{ var}}$$

$$\frac{dJ}{da} = \frac{dJ}{dv} \frac{dv}{da}$$

$$\boxed{\frac{dJ}{da} = 3}$$

$$a = 5$$

$$u = bc$$

$$v = a + u$$

$$b = 3$$

$$\boxed{dv = 3}$$

$$c = 2$$

$$u = 6 \rightarrow 6.001$$

$$v = 11 \rightarrow 11.001$$

$$J = 33 \rightarrow 33.003$$

$$J = 34$$

$$\frac{dJ}{du} = 3 = \frac{dJ}{dv} \frac{dv}{du}$$

$$\frac{dJ}{dv} =$$

$$= 3 \times 1$$

$$\frac{dJ}{db} = \frac{dJ}{du} \frac{du}{db}$$

$$b = 3 \rightarrow 3.001$$

$$\frac{dJ}{db} = \frac{dJ}{du} \frac{du}{db}$$

$$= 3 \times 2$$

$$= 6$$

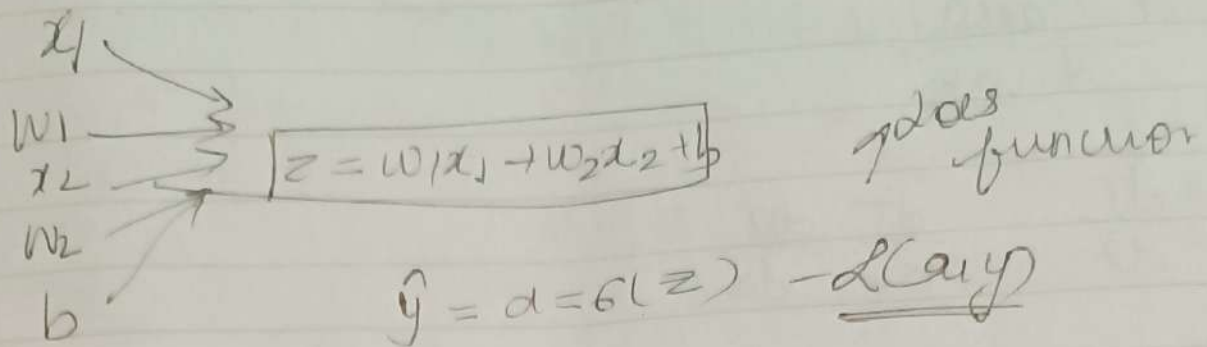
$$\boxed{Z = W^T x + b}$$

Logistic Regression

$$\boxed{Z = W^T x + b}$$

$$\boxed{\hat{y} = \sigma(z)}$$

$$L(a, y) = -(y \log(a) + (1-y) \log(1-a))$$



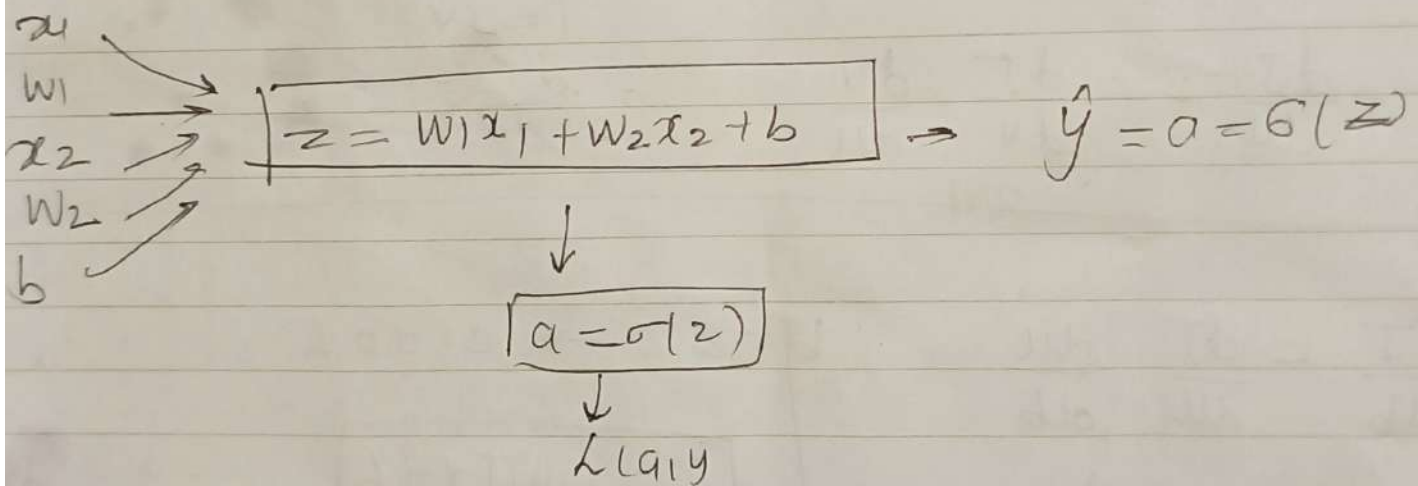
$$\hat{y} = a = \sigma(z) = \text{sigmoid}(z)$$

We will take derivative of loss function with respect to w

$$z = w^T x + b$$

$$\hat{y} = a = \sigma(z)$$

$$\mathcal{L}(a|y) = (y \log(a) + (1-y) \log(1-a))$$



$$\frac{da}{dz} = \frac{d\mathcal{L}(a|y)}{da} = -\frac{y}{a} + \frac{1-y}{1-a}$$

$$\frac{d\mathcal{L}}{dz} = \frac{d\mathcal{L}}{da} \cdot \frac{da}{dz}$$

$$= a - y$$

Gradient Descent on m example

$$J(w, b) = \frac{1}{m} \sum L(a^{(i)}, y^{(i)})$$

$$a^{(i)} = y^{(i)} = \sigma(z^{(i)}) = \sigma(w^T x^{(i)} + b)$$

$$dw_1^{(i)}, dw_2^{(i)}, db^{(i)}$$

$$\frac{\partial J(w, b)}{\partial w_1} = \frac{1}{m} \sum_{i=1}^m \frac{\partial L(a^{(i)}, y^{(i)})}{\partial w_1} = \frac{1}{m} \sum_{i=1}^m (x^{(i)} - y^{(i)})$$

$$J=0 \quad dw_1=0 \quad dw_2=0 \quad db=0$$

For $i=1$ to m

$$z^{(i)} = w^T x^{(i)} + b$$

$$a^{(i)} = \sigma(z^{(i)})$$

~~$$z^{(i)} = w^T x^{(i)} + b$$~~

$$dz^{(i)} = a^{(i)} - y^{(i)}$$

$$dw_1 += x_1^{(i)} dz^{(i)}$$

$$dw_2 += x_2^{(i)} dz^{(i)}$$

$$db += dz^{(i)}$$

$$J /= m$$

$$dw_1 /= m, \quad dw_2 /= m, \quad db /= m$$

$$J = -y^{(i)} \log a^{(i)} - (1 - y^{(i)}) \log (1 - a^{(i)})$$

$$w_1 = w_1 - \alpha dw_1$$

$$w_2 = w_2 - \alpha dw_2$$

$$b = b - \alpha db$$

1st for loop for all example
and second for loop
for feature